Federal Economist Compensation & PhD Placement: Evidence by Gender

Updated Data and New Sources

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Outline

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2. Federal Salary Analysis
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Overview

This project extends prior work on gender equity among economists in two ways:

1. Updates the federal salary analysis through FY2024 using FedsDataCenter (previously through FY2015 in Foster, Manzella, McEntarfer & Sandler 2020)
2. Introduces a new data source on PhD economist initial placements (44 programs, 1991–2025) to complement the LEHD-based analysis in Foster, McEntarfer & Sandler (2023)

Key questions:

- How has the federal economist gender pay gap evolved since 2015?
- How do initial placement shares compare across sectors and over time?
- How does the gender composition of placements compare to the full SED/LEHD sample?

Data Sources

Federal salary data

- Source: FedsDataCenter.com (OPM data via FOIA)
- Years: FY2015–FY2024
- ≈4,300–4,700 economists per year
- Occupation code: "Economist"
- Fields: name, agency, grade, pay plan, salary, bonus
- Gender assigned via `gender_guesser` (offline, ≈84% coverage)

PhD placement data

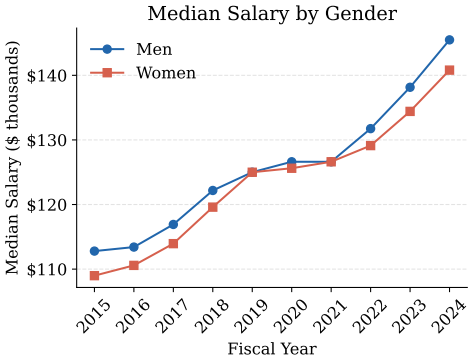
- Source: econphdplacements.com
- 8,824 placements from 44 programs
- Includes US and international programs (e.g., Stockholm, UBC, LSE)
- Years: 1991–2025
- Fields: name, year, program, field, placement, category
- Gender assigned via `gender_guesser` (≈51% coverage)

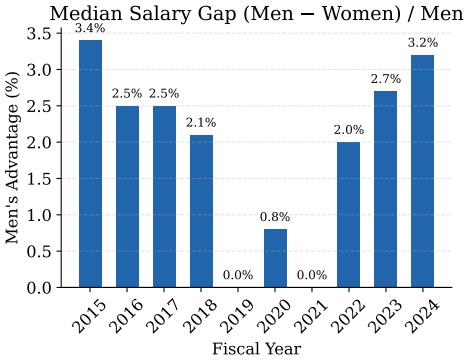
A Note on Gender Assignment

Both datasets use **name-based gender assignment** via `gender_guesser` (an offline library covering ≈40,000 names):

- Returns: *male*, *mostly_male*, *female*, *mostly_female*, *androgynous*, *unknown*
- We assign male/female from the first two; leave the rest unassigned
- Salary data: **84%** coverage (OPM names are mostly Anglo-American)
- Placement data: **51%** coverage (many international names not in library)
- Unassigned names skew international and likely male-heavy ⇒ female shares in placement data are *understated*
- Prior papers (Foster et al. 2020, 2023) use **self-reported** SED gender (100% coverage, no bias)

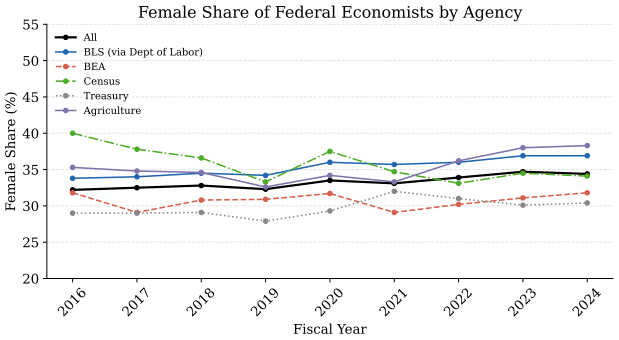
Median Salary by Gender, FY2015–2024





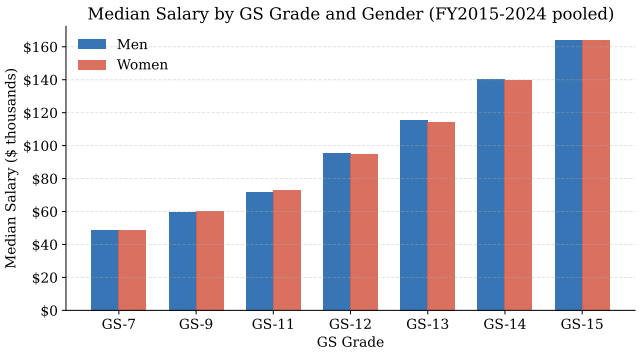
- Gap peaked at 3.4% in 2015, nearly closed in 2019–2021, then **re-widened to 3.2%** by 2024
- 2024 medians: men $145,489 women $140,807

Female Share of Federal Economists by Agency



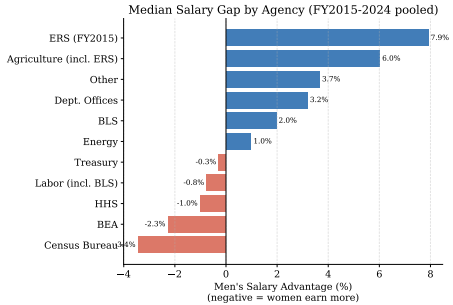
- Overall share rising: 32% (2016) $\rightarrow$ 34% (2024)
- HHS consistently highest ($\approx$40–45%); Treasury lowest ($\approx$29–32%)

Is the Gap Within or Between Grade?



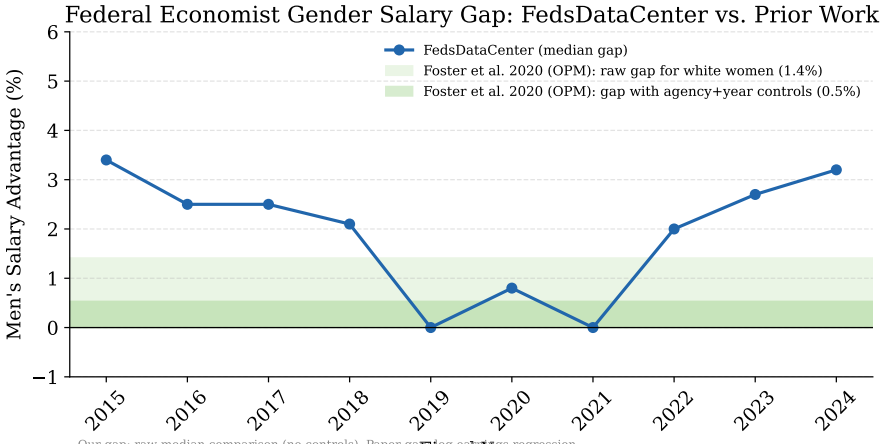
- Within-grade gaps are **near zero or reversed** across all GS levels
- The overall salary gap is driven by **grade distribution**: women are concentrated at lower GS grades

Salary Gap by Agency



- BEA, Census Bureau, and BLS show female salary advantage (women earn more at median)
- Largest male advantage: Agriculture/ERS and "Other" agencies
- Pattern consistent with within-grade gaps being near zero: agency differences reflect grade composition

Federal Salary Gap: FedsDataCenter vs. Foster et al. (2020)



Federal Salary Gap: Comparison Summary

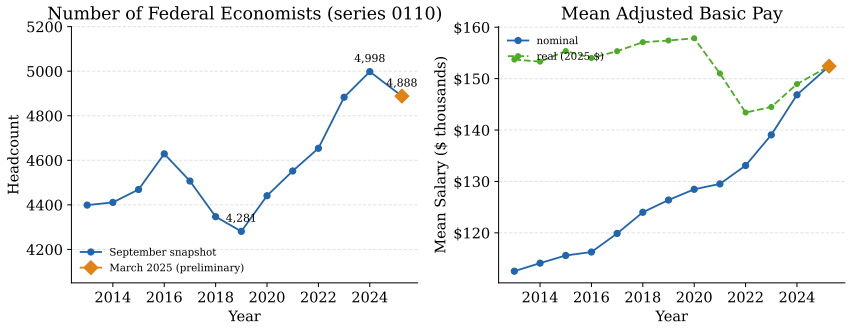
Source	Sample	Specification	Gap
FedsDataCenter (FY2015)	All GS economists	Raw median	3.4%
FedsDataCenter (FY2024)	All GS economists	Raw median	3.2%
FedsDataCenter (FY2019–21)	All GS economists	Raw median	≈0%
Foster et al. (2020)	PhD economists, OPM	Raw log gap (white women)	1.4%
Foster et al. (2020)	PhD economists, OPM	Agency + year FE	0.5%
Foster et al. (2023)	PhD economists, LEHD	Raw log gap, government	8.6%
Foster et al. (2023)	PhD economists, LEHD	Full controls, government	3.8%

- Our raw 3% median gap vs. OPM-based 1.4% raw log gap: broadly consistent
- LEHD-based 8.6% is larger because it is **PhD-only** and captures broader compensation (not just base salary)

The 2025 Update: Data

- Question: has the **number** and **pay** of federal economists changed into 2025?
- FedsDataCenter stops at FY2024. 2025 uses **OPM FedScope**: preliminary **March 2025** snapshot with **September 2024** bundled for comparison (occupational series 0110).
- Three caveats up front:
  - **No gender / race** — FedScope's raw files never had them (the interactive site did, until removed in March 2025). This update is headcount and pay only.
  - **Preliminary**, and still counts administrative-leave / deferred-resignation staff as employed $\Rightarrow$ headcount loss understated.
  - Different universe from FedsDataCenter (EHRI month-end status); small agency $\times$ grade cells suppressed ($\sim$76% coverage).

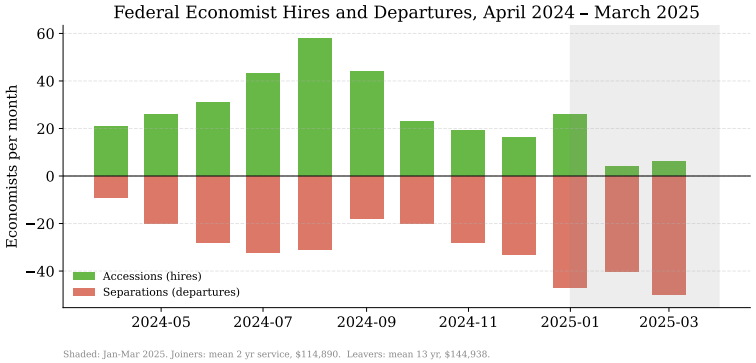
Number and Pay of Federal Economists



Source: OPM FedScope employment cubes (Sep) and the preliminary March 2025 snapshot — one consistent universe. March 2025 still counts administrative-leave / deferred-resignation staff as employed. Real pay deflated by CPI-U to March 2025 dollars.

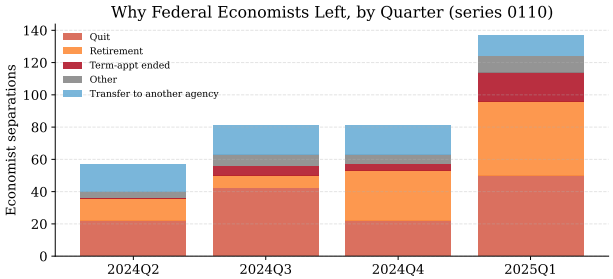
- Headcount −2.2% (4,998 → 4,888), Sep 2024 → Mar 2025.
- Mean pay +2.8% nominal, +1.4% **real** (CPI-U) — almost entirely the January 2025 pay raise (2.0%), not a shift to more senior staff.

Hires and Departures: the Clearer Signal



- Quarterly net: +21, +64, −23, −101 (2025 Q1). Hiring collapsed (145 → 36); departures spiked (81 → 137).
- Against normal Q1 turnover the excess is a **tenure barbell**: staff under 2 years and with 20+ years — not mid-career economists.

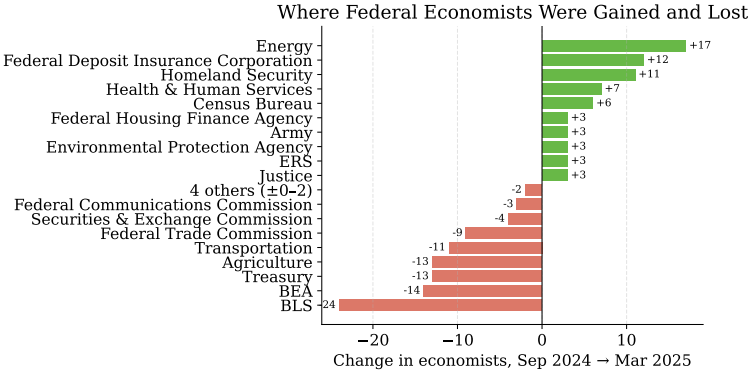
Why They Left: A Drawdown, Not Ordinary Turnover



2025 Q1 holds 86% of the year's early-out retirements, 64% of term-appointment endings, 42% of voluntary retirements and 37% of quits; transfers to other agencies fell — economists left government rather than moving within it.

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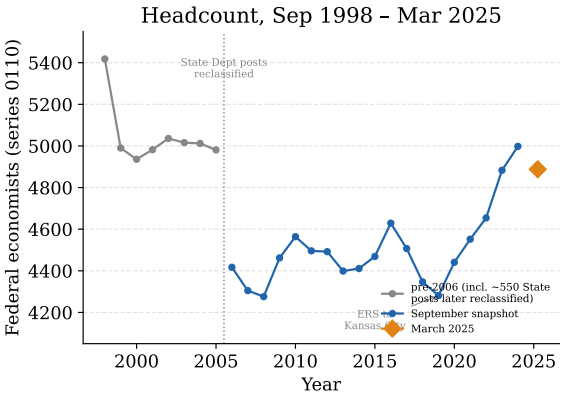
Where Economists Were Gained and Lost

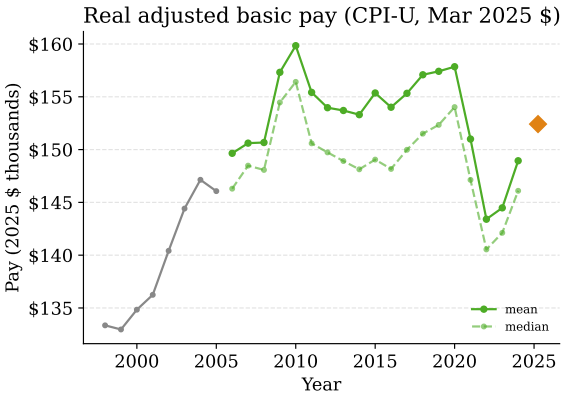


Agency × pay-plan × grade cells with ≤10 staff are suppressed (~76% coverage). Agencies changing by 0–2 are pooled.

- Statistical agencies among the losers: **BLS** −24**, BEA** −14; also Treasury −13, Agriculture −13, FTC −9.
- Gained: Energy +17, FDIC +12, DHS +11, HHS +7, Census +6.

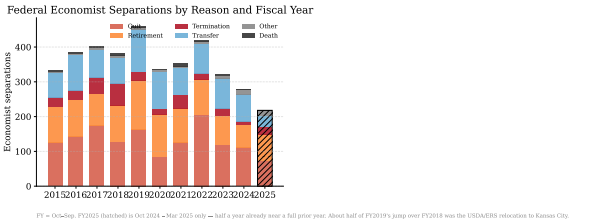
Historical Context: Economists Since 1998

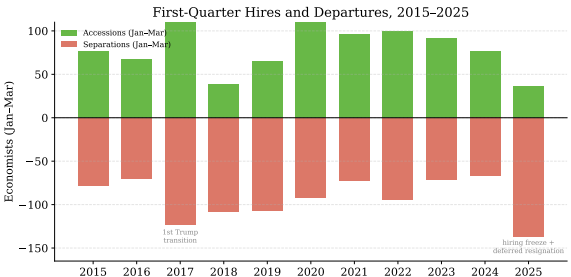




- On a consistent post-2006 basis, headcount ran 4,280–5,000. March 2025 (4,888) is still near the *top* of that range — the drop so far unwinds about a year of growth, not decades.
- Real pay is mid-range: above the late 1990s, below the 2010 and 2018–20 peaks.

Historical Context: 2025 Outflow vs. Prior Years





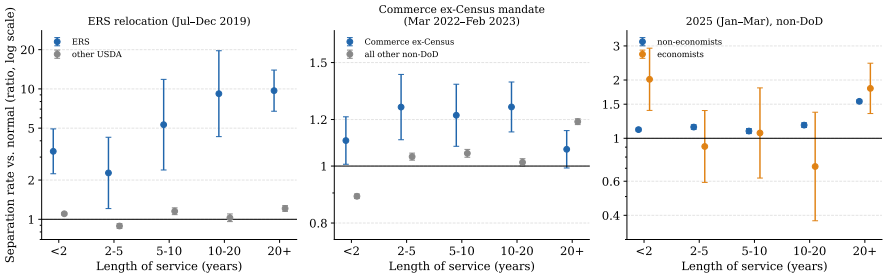
- FY2024 was the calmest year in the series (279 separations); FY2025's **first six months** already hit 218. Jan–Mar 2025 had the most departures (137) and the fewest hires (36) of any Q1 since 2015.
- The one precedent — FY2019 — was about half one agency (the ERS move to Kansas City). Formal RIFs are near-zero for economists in every year, so 2025's exits run through quits, retirements and term endings. The return-to-office order and the buyout offer are co-timed.

2025 Update: Robustness

Definition	Sep 2024	Mar 2025	Headcount	Pay
0110 economist (headline)	4,998	4,888	−2.2%	+2.8%
0110 + 0119 economics assistant	5,333	5,219	−2.1%	+2.8%
0110 with a doctorate (Foster proxy)	—	1,386	—	mean $179k

- Result is not sensitive to the series definition.
- Foster et al. (2020, 2023) restrict to **PhD** federal economists — ≈ the 1,386 doctorate holders here (28% of 0110). Explains their Treasury/Agriculture/HHS focus vs. our BLS-heavy full population.
- Shared universe exclusions: CEA, Federal Reserve Board, CBO, USPS, intelligence agencies.

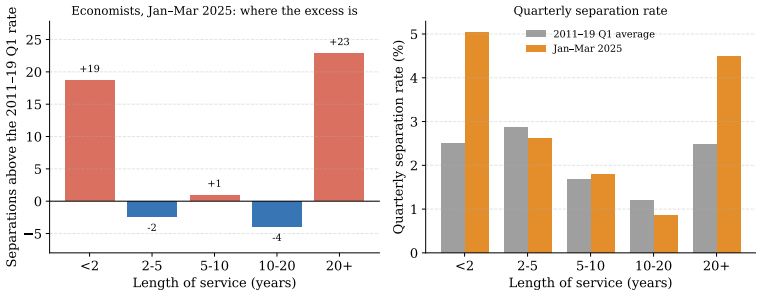
Who Leaves When Workplace Conditions Change?



Ratio of the window's separation rate to the same window in 2011–18/2011–19, 95% Poisson CIs. DoD excluded (tenure redacted). Tenure of stock is measured at the prior September.

- **ERS move (2019):** separations 3.6× normal, ≈ 10× at 10+ years of service. **Commerce mandate (2022):** no clear effect.
- **2025 (non-DoD):** +25%, 1.55× at 20+ years; economists a barbell (under 2 yrs, 20+ yrs). DoD excluded: tenure is redacted.

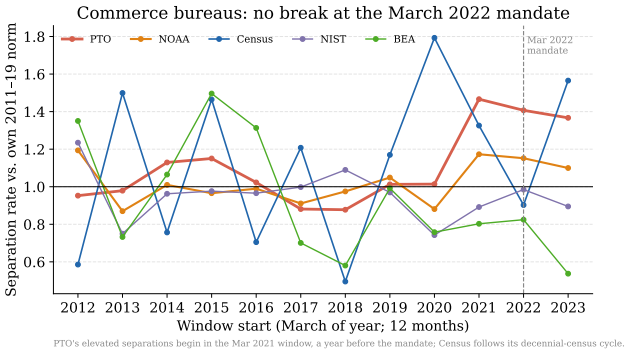
Economists in 2025: Excess Exits at Both Ends of Tenure



Non-DoD economists (n=137 separations; DoD occupations are redacted). Excess is rate-adjusted for workforce growth.
Junior excess: quits, terminations, other; senior excess: retirements, early-outs.

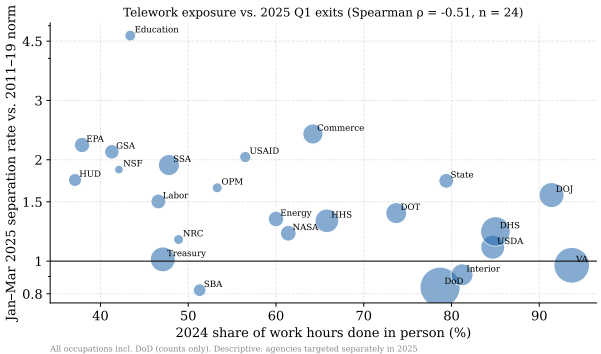
- 137 separations vs. 99 expected at the 2011–19 Q1 rate (non-DoD): +19 under 2 years (quits, terminations), +23 at 20+ years (retirements).
- Mid-tenure (2–20 years) economists left no more than usual.

Commerce, March 2022: No Break at the Mandate



- Department-level difference-in-differences +11%, but 2012 and 2023 windows are as large; economists show no effect.
- PTO's elevated separations begin in the March 2021 window — a year *before* the mandate.

Telework Exposure and 2025 Exits



- Agencies with less in-person work saw larger Q1 increases (Spearman $\rho = -0.51$, $n = 24$)...
- ...but agencies targeted for separate reductions (e.g., Education, USAID) are also low in-person: the return-to-office order and the buyout cannot be separated.

PhD Economist Placement Data: Overview

Sample characteristics

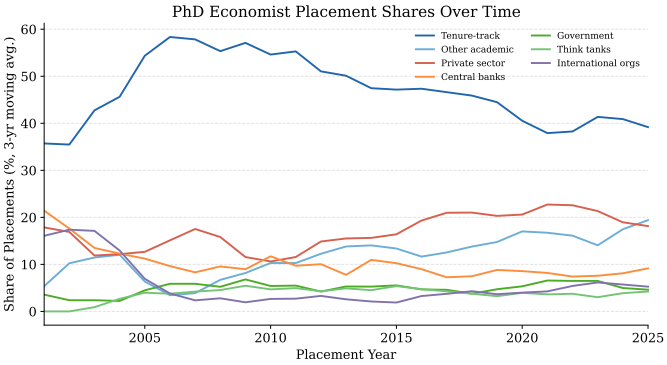
- 8,824 placements, 44 programs
- Placement years: 1991–2025
- Includes international programs: Stockholm, UBC, LSE, Sciences Po, Mannheim, EUI, CEMFI, ...
- 7 placement categories: tenure-track, other academic, private sector, central banks, government, think tanks, international orgs

Overall placement distribution

Category	Share
Tenure-track	49.4%
Private sector	18.8%
Other academic	12.2%
Central banks	7.2%
International orgs	4.6%
Think tanks	3.8%
Government	4.1%

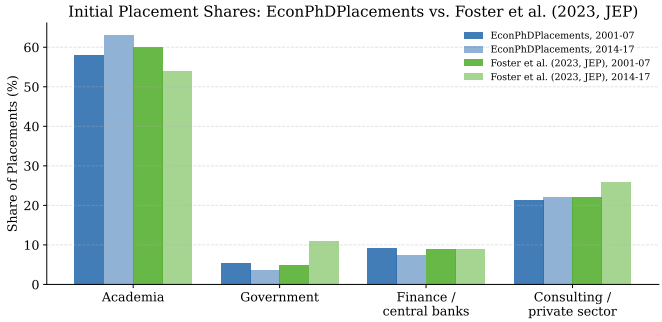
Academia total: 61.6%

Placement Trends Over Time



- Tenure-track share declining since ≈2010; other academic rising
- Private sector growing steadily; central bank placements relatively stable

Placement Shares: EconPhDPlacements vs. Foster et al. (2023, JEP)



Note: 'International orgs' (4-6% in EconPhDPlacements) not captured in paper (LEHD = US only).
Paper: Foster, McEntarfer & Sandler (2023, JEP). Govt in paper = NAICS 92 only.

Reconciling the Differences

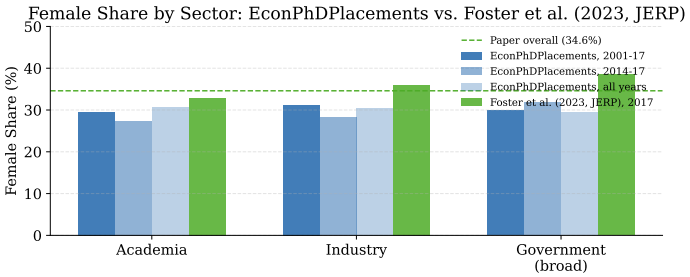
Our academia share (63%, 2014–17) is higher than paper's (54%, all programs):

- Our 44 programs skew toward **top research universities** — even for top-5 programs, paper shows 58.6%
- Placement website data = **announced** initial placements; LEHD = actual employment (captures people who planned academia but moved)
- International programs in our sample (Stockholm, UBC, ...) may have higher academic placement rates

Our government share (3.6%, 2014–17) is lower than paper's (11%):

- Top programs go to government less (paper's top-5: 8.6%; others: 11.4%)
- Government placements may be under-reported on department websites
- Paper's government includes all levels (federal + state/local, NAICS 92)

Female Share by Sector: EconPhDPlacements vs. Foster et al.



Note: Gender assigned via gender_guesser (50.5% of placements). Unassigned names skew international (likely male-heavy), so our female shares are likely understated. Paper uses self-reported SED gender (100% coverage). D&E = Diversity & Equity in Labor Market Outcomes for Economists (Foster, McEntarfer & Sandler, 2023, JERP).

Female Share in Placements: Caveats and Takeaways

Our estimates (29–32%) are below the paper's (33–38%):

- **Gender assignment gap:** only 51% assigned; unassigned names are disproportionately international and likely male-heavy $\Rightarrow$ our female shares are understated by an unknown amount
- **Initial placements vs. cross-section:** paper shows the 2017 stock of employed PhD economists (includes earlier cohorts with higher female retention); ours shows the flow of new placements
- **Ranking within sectors is preserved:** government $>$ industry $>$ academia in both datasets

Next step: use gender-specific name databases (e.g., Social Security Administration data, genderize.io) to improve coverage for international names

Summary I

Federal salary (FedsDataCenter, FY2015–2024):

- Raw median gender gap: 0–3.4%, broadly consistent with Foster et al. (2020)
- Gap is entirely between-grade, not within-grade; BEA, Census, BLS show female salary *advantage* at median
- Female share rising: 32% $\rightarrow$ 34%; gap re-widened post-2021

The 2025 update (OPM FedScope, Sep 2024 $\rightarrow$ Mar 2025):

- Headcount -2.2%; mean pay $+2.8\%$ nominal / $+1.4\%$ real (almost entirely the January 2025 raise)
- Flows show the turn: 2025 Q1 net -101, hiring collapsing; the excess exits are staff under 2 years and with 20+ years of service
- BLS and BEA among the agencies losing economists

Summary II

- No gender in FedScope's raw files (any year); FedsDataCenter, with name-based gender, ends at FY2024

Who leaves when workplace conditions change (three episodes):

- ERS relocation (2019): clear, senior-skewed exodus; Commerce 2022 mandate: no mandate-timed break; 2025: broad surge (+25%) that the return-to-office order and buyout offer jointly explain

PhD placements (econphdplacements.com, 1991–2025):

- Sample skews toward top programs; academia share higher than national average
- Government placements under-represented vs. LEHD-based paper
- International orgs (4–6%) visible here but absent from LEHD data
- Female share understated due to gender assignment coverage; ordering across sectors preserved

Summary III

Next steps:

- Improve gender assignment for international names
- Link placement and salary data on individual economists
- Extend to regression-based gap estimates controlling for grade/agency

References I

- Foster, L., Manzella, J., McEntarfer, E., & Sandler, D.H. (2020). "Employment and Earnings for Federal Government Economists: Empirical Evidence by Gender and Race." *AEA Papers and Proceedings*, 110, 210–214.
- Foster, L., McEntarfer, E., & Sandler, D.H. (2023). "Diversity and Equity in Labor Market Outcomes for Economists." *Journal of Economics, Race, and Policy*.
- Foster, L., McEntarfer, E., & Sandler, D.H. (2023). "Early Career Paths of Economists Inside and Outside of Academia." *Journal of Economic Perspectives*, 37(4), 231–250.
- Data: FedsDataCenter.com (OPM federal salary data, FY2015–2024)
- Data: econphdplacements.com (PhD economist placement data, 1991–2025)